Nonparametric Statistics Skill — Overview

File structure and what each file does

Hsieh Po Sheng

Department of Statistics, National Cheng Kung University

2026/10/15

Outline I

1. File Structure
2. SKILL.md
3. methods.md
4. textbook-methods.md
5. decision-tree.md
6. run_test.py

Section 1

File Structure

File Structure

The skill uses **progressive disclosure**: a lightweight main file that is always loaded, detailed knowledge split under `references/`, and computation delegated to `scripts/`.

```
nonparametric-stats-skill/
   SKILL.md                    <- main file: behavior & workflow
   references/
       methods.md              <- quick reference (scipy + effect sizes)
       textbook-methods.md     <- textbook detail (assumptions, formulas)
       decision-tree.md        <- method-selection tree (Phase 0 + main tree)
   scripts/
       run_test.py             <- actual computation + plotting
```

- **SKILL.md** — always loaded; controls *how the agent behaves*.
- **references/** — looked up on demand; holds the *knowledge*.
- **scripts/** — run only when numbers must be computed or data plotted.

One-line Summary

File	Purpose
`SKILL.md`	Defines behavior, workflow, triggers (the brain)
`methods.md`	One-page cheat sheet for the 6 tests
`textbook-methods.md`	Full assumptions & formulas (deepest layer)
`decision-tree.md`	Helps pick *which test* to use
`run_test.py`	Actually computes (scipy) and plots (matplotlib)

Scope matches this project's `slides.qmd` (based on Hollander, Wolfe & Chicken, 2015): ***6 methods****.*

Section 2

SKILL.md

SKILL.md — the Brain

What it does: defines the skill's trigger conditions, scope, and the agent's **behavior and answering workflow**. Editing this file changes how the agent acts.

- **Behavior guardrails (top priority):**
  - After data is imported, **ask before plotting** (even a scatter plot).
  - **Never run any test without explicit user consent.**
  - Do one step at a time, then stop and wait for the user.
- **Answering workflow:** clarify key points -> describe data & ask about a scatter plot -> select method (ask before running) -> assumptions & details -> effect sizes -> parametric vs. nonparametric trade-off.

  Core principle: always ask, never act on its own.

Section 3

methods.md

methods.md — Quick-Reference Card

What it does: a one-page cheat sheet for the 6 methods, aimed at **computation**.

Each method lists:

- Null hypothesis H_0
- Test statistic
- Handling of ties
- Effect size
- The matching **scipy function** (e.g. `scipy.stats.mannwhitneyu`)

  Use this to quickly look up "how is this test computed, which scipy function, what effect size."

Section 4

textbook-methods.md

textbook-methods.md — Textbook-Level Detail

What it does: the deepest theoretical layer, extracted from `slides.qmd`.

Each method includes:

- Full assumptions (A1 / A2 / A3)
- Test statistic and derivation
- Tie-corrected variance formulas
- Exact decision rules and large-sample normal approximation
- R code

  Covers Wilcoxon signed-rank, Mann-Whitney, Ansari-Bradley, two-sample KS, Kruskal-Wallis, Friedman. The place for formulas and rigorous justification.

Section 5

decision-tree.md

decision-tree.md — Phase 0: Understand the Data

Before choosing a method: fix the **sample structure**, then check assumptions **graphically** (normality, symmetry, similar shape).



decision-tree.md — Method-Selection Tree

Structural branching: purpose -> sample structure -> method (6 methods only).

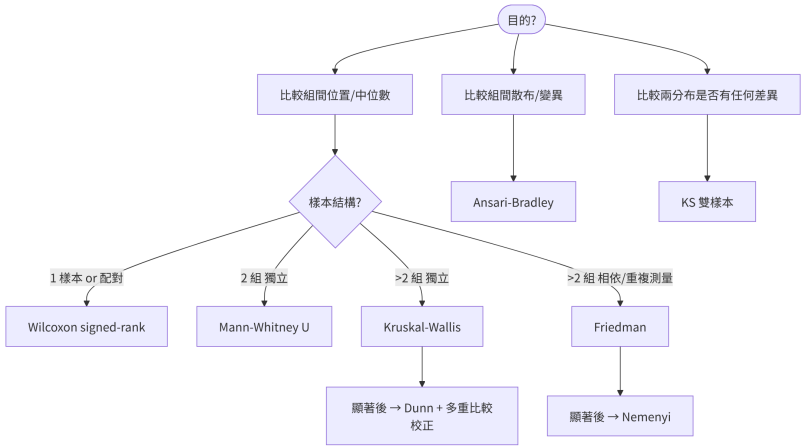


Figure 2: Main tree: 6 methods only

Section 6

run_test.py

run_test.py — Compute + Plot

What it does: SKILL.md requires "always run it, never compute by hand" — this is the script that runs.

- **Tests** (scipy; returns statistic, p-value, effect size):
  - `mannwhitney`, `wilcoxon`, `kruskal`
- **Plotting** `plot` (matplotlib; saves PNG):
  - `scatter` — scatter plot
  - `box` — boxplot + jittered points
  - `paired` — before/after + difference distribution
  - `hist` — histogram + normal curve + Q-Q

```
python scripts/run_test.py plot --kind scatter --csv data.csv --x h --y w
python scripts/run_test.py mannwhitney --csv data.csv --value score --group g
```

Summary

- **Design:** progressive disclosure (light main file + layered references + computation script).
- **Behavior:** always ask; no plotting or testing without consent.
- **Scope:** aligned to the 6 nonparametric methods in `slides.qmd` — all four files and the script are consistent.